8 Queen Problem

Aim

To solve the **8-Queen Problem** using **Backtracking** in Python such that no two queens attack each other.

Algorithm

1. Start from the first column.
2. Place a queen in each row one by one.
3. Check whether the queen is safe:
    - No other queen in the same row.
    - No queen in the upper-left diagonal.
    - No queen in the lower-left diagonal.
4. If safe, place the queen and move to the next column.
5. If no position is safe, backtrack and try another row.
6. Repeat until all 8 queens are placed.

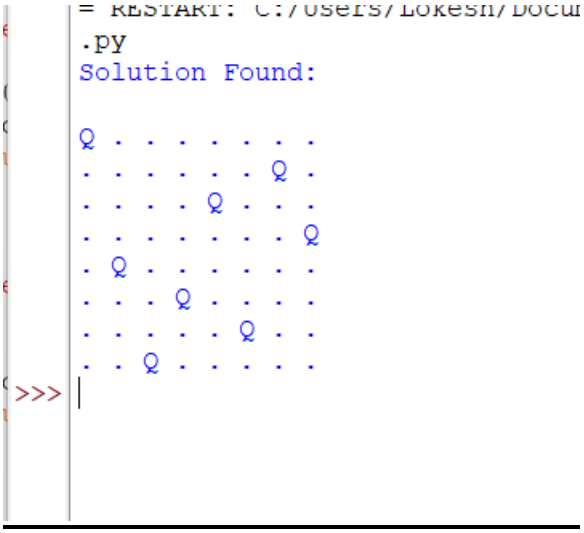
Python Code

```
N = 8


def print_board(board):
    for row in board:
        print(" ".join("Q" if x == 1 else "." for x in row))


def is_safe(board, row, col):
    # Check left side of row
    for i in range(col):
        if board[row][i] == 1:
            return False

    # Check upper-left diagonal
    i, j = row, col
```

```
    while i >= 0 and j >= 0:
        if board[i][j] == 1:
            return False
        i -= 1
        j -= 1

    # Check lower-left diagonal
    i, j = row, col
    while i < N and j >= 0:
        if board[i][j] == 1:
            return False
        i += 1
        j -= 1

    return True

def solve(board, col):
    if col >= N:
        return True

    for i in range(N):
        if is_safe(board, i, col):
            board[i][col] = 1

            if solve(board, col + 1):
                return True

            board[i][col] = 0
```

```
    return False


board = [[0 for _ in range(N)] for _ in range(N)]


if solve(board, 0):
    print("Solution Found:\n")
    print_board(board)
else:
    print("No Solution Exists")
```

Output



Result

Thus, The 8-Queen Problem was successfully solved using the **Backtracking Algorithm** in Python.